1. Quadratics

1.1. Quadratic Formula

The quadratic formula helps us find the solutions to a quadratic equation, $ax^2+bx+c=0$.

It helps us find the values of x for which the quadratic equation equals 0, which are also called the zeros of the quadratic equation for this reason.

Quadratic formula (Given in MF19)

$$x=\frac{-(b)\pm\sqrt{(b)^2-4(a)(c)}}{2a}$$

So each of the roots of the quadratic equation $ax^2+bx+c=0$ are

$$x=\frac{-b+\sqrt{b^2-4ac}}{2a}$$

And

$$x=\frac{-b-\sqrt{b^2-4ac}}{2a}$$

Factorisation

Factorisation allows us to express a polynomial as the **product** of simpler terms.

For example:

$$x^2-2x-x+2=(x-1)(x-2)$$

Factorisation of a quadratic can be done through two methods:

- splitting the middle term
  - Re-writing $ax^2+bx+c=0$ as $ax^2+dx+ex+c=0$ [Where $d\times e=a\times c$ and $dx+ex=bx$]
  - Then factoring out the common factors in the equation
- A calculator
  - Using the "mode" button and then selecting the type of equation. This will give us the solutions to a quadratic equation, which can then be used to factorise it.

Example 1

Solve the quadratic equation $6x^2-17x+5=0$ using the quadratic formula.

Answer

Here $a=6,b=-17,c=5$. Using the quadratic formula

$$x=\frac{-(-17)\pm\sqrt{(-17)^2-4(6)(5)}}{2(6)}$$

So the 2 values of x which satisfy the equation are

$$x=\frac{5}{2}\text{ and }x=\frac{1}{3}$$

Example 2

Solve the quadratic equation $6x^2-17x+5=0$ using factorisation.

Answer

This can be factorised as

$$(2x-5)(3x-1)=0$$

- Logically, if we have $ab=0$ then either $a=0$ or $b=0$.

$$2x-5=0\implies x=\frac{5}{2}$$

$$3x-1=0\implies x=\frac{1}{3}$$

1.2. Completing the square

Completing the square allows us to write a quadratic, ax^2+bx+c , in the form $p(x+q)^2+r$. This is useful for a variety of reasons, which we will see later on.

- Given ax^2+bx+c , we can complete the square using the formula:

$$a\left(x+\frac{b}{2a}\right)^2+c-\frac{b^2}{4a}$$

- Comparing coefficients in the expansion of $(x+a)^2$ is another way to complete the square.

Example

Express $3x^2+9x+5$ in the form of $p(x+q)^2+r$, where p, q and r are constants.

Answer

First we identify our coefficients, $a=3,b=9,c=5$. Now we can substitute in our values into the formula:

$$3\left(x+\frac{9}{2(3)}\right)^2+5-\frac{9^2}{4(3)}$$

Which simplifies to

$$3\left(x+\frac{3}{2}\right)^2-\frac{7}{4}$$

- You can verify your answer by expanding it and comparing it with your original quadratic.

1.3. Solving quadratic equations using completing the square form

It is possible to solve quadratic equations by completing the square. Given a quadratic equation $ax^2+bx+c=0$, it can be written as $p(x-q)^2+r=0$.

- By making x the subject of the formula, we get

$$x=\pm\sqrt{\frac{-r}{p}}+q$$

- Note that if $\frac{-r}{q}<0$, the quadratic has no real solutions.

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Example

By completing the square, find the solutions to the quadratic equation $x^2 - 5x + 6 = 0$.

Answer

- First we complete the square

$$x^2 - 5x + 6 \implies \left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$$

$$\left(x - \frac{5}{2}\right)^2 - \frac{1}{4} = 0$$

- Next, we make x the subject of the formula and solve for it

$$x = \pm\sqrt{\frac{1}{4}} + \frac{5}{2}$$

$$x = 3 \quad \text{or} \quad x = 2$$

1.4. Finding the coordinates of the vertex

The vertex of a quadratic equation is the **highest** or **lowest** point on the curve. We can find the coordinates of the vertex after completing the square

Equation	Coordinates of vertex
$y = p(x-q)^2 + r$	(q, r)
$y = p(x+q)^2 - r$	$(-q, -r)$

Another way of finding coordinates of the vertex is through the general completing the square formula. Given a quadratic, $y = ax^2 + bx + c$:

$$y = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

x coordinate of vertex	$x = \frac{-b}{2a}$
y coordinate of vertex	$y = \frac{4ac - b^2}{4a}$

- The line $x = \frac{-b}{2a}$ acts as a **line of symmetry** for the quadratic. It splits the quadratic curve into 2 equal and mirrored parts.
- Completing the square is also called the vertex form due to this.

Example

Find the coordinates of the vertex of the quadratic $3\left(x + \frac{3}{2}\right)^2 - \frac{7}{4}$.

Answer

- We observe that it has been written in the completing the square, or vertex, form.

$$3\left(x + \frac{3}{2}\right)^2 - \frac{7}{4} = p(x-q)^2 + r$$

$$q = -\frac{3}{2}$$

$$r = -\frac{7}{4}$$

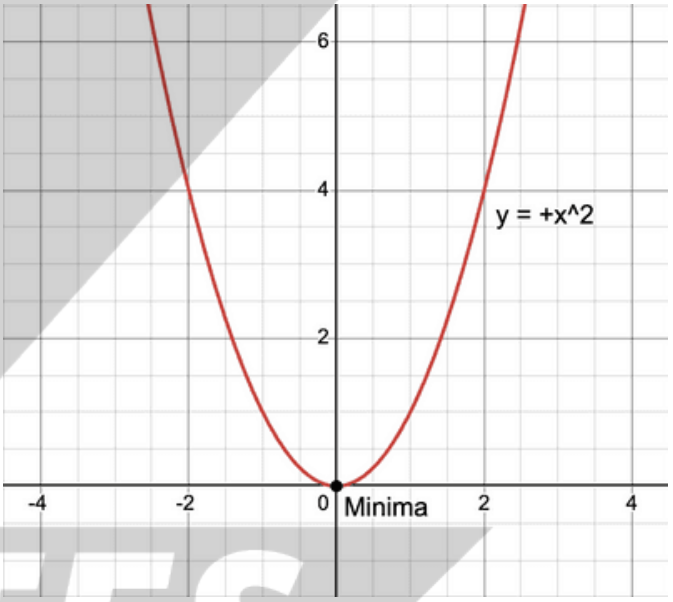
$$\text{The coordinates of the vertex are } \left(-\frac{3}{2}, -\frac{7}{4}\right)$$

1.5. Quadratic inequalities

An inequality question will ask us to find the values of x for which the inequality holds true.

The most efficient way to solve these inequalities is to follow these steps.

- Factorise the inequality.
  - This will help us find the **intersections** of the quadratic with the x-axis
- Draw a **simple** sketch of the quadratic and label the x coordinate of the intersections.
  - The quadratic curve will have a "**smiley**" or an **U shape** if the **coefficient** of x^2 is **positive.** This point is also known as a **minimum point**, or a **minima**.



  - The quadratic curve will have a "**frown**" or a **n shape** if the **coefficient** of x^2 is **negative.**

    This point is also known as a **maximum point**, or a **maxima**.

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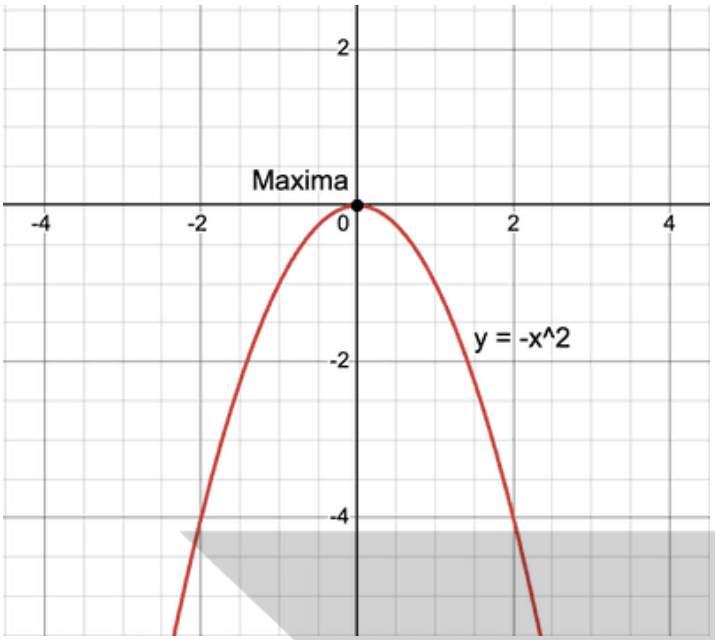


- Using your sketch, deduce the values of x for which the inequality holds true.
- If a question asks you to find $y \leq 0$ or $y \geq 0$, then the points of intersections must be included in the range of values.

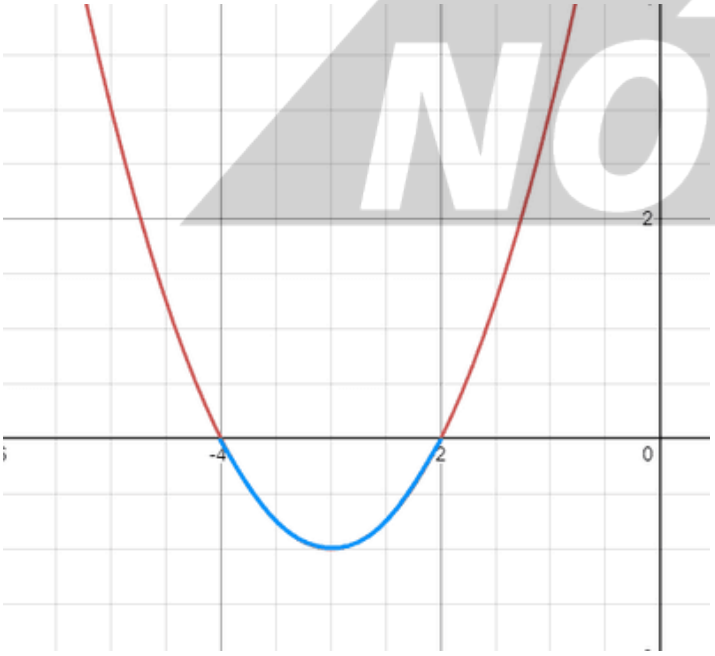
Example

Find the values of x which satisfy the inequality $x^2 + 6x + 8 < 0$

Answer

- First notice that the coefficient of the quadratic is positive, $+x^2$, meaning the curve must be sketched in the shape of a smiley or an u.
- Then we can factorise $x^2 + 6x + 8$ to get $(x+4)(x+2)$. Upon factoring, it is clear that the roots are $x = -2$ and $x = -4$.

We can now sketch the parabola $y = (x+4)(x+2)$, labelling the intersections with the x-axis



The blue part of the curve represents the region where $y < 0$ and the red part of the curve represents the region where $y > 0$

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- As we're trying to find the values of x for which $y < 0$, we look at the blue region. Clearly any value that is greater than -4 and less than -2 is represented in the blue region. So the **answer** is $-4 < x < -2$.

1.6. Discriminant of a quadratic formula

The discriminant helps us find the **number of intersections** of a **quadratic** equation with the x-**axis**. It can also be used to find the **number of intersections** of a **quadratic curve** with a **line** or **another quadratic curve**.

Consider $b^2 - 4ac$ in quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- If $b^2 - 4ac > 0$
  - This results in $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$, meaning there are **two distinct real roots** that can be found.
- If $b^2 - 4ac = 0$
  - As the square root is **equal to 0**, we get $x = \frac{-b}{2a}$, there are **two equal** and **real roots.**
- If $b^2 - 4ac < 0$
  - There is a **negative** term inside the square root, later taught as a complex root, meaning there are **no real solutions**.

  For finding the number of intersections with the x-axis, or finding the **number of roots**:

Discriminant	Number of roots	Nature of intersection with x-axis
$b^2 - 4ac > 0$	Two real distinct roots	Two distinct points of intersection
$b^2 - 4ac = 0$	Two equal real roots	One point of intersection
$b^2 - 4ac < 0$	No real roots	No points of intersection

Finding the number of intersection between curves

- The theory of discriminants can be extended to find the number of intersections between a quadratic and another quadratic curve or linear line.
  - To find the number of intersections, we first **equate both curves** or line and then apply the discriminant formula.

    For finding the number of intersections between a **quadratic** and **another** line or curve:

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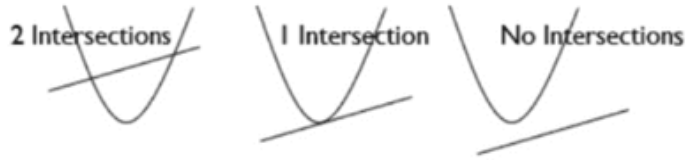
Discriminant	Number of intersections	Nature of intersection with curve
$b^2 - 4ac > 0$	Two real distinct points of intersection	Line meets curve at two points
$b^2 - 4ac = 0$	One real point of intersection	Line is **tangent** to the curve
$b^2 - 4ac < 0$	No real points of intersections	Line does not meet the curve



Example

Given a quadratic, $kx^2 + 4kx + 3k = 0$, find the range of k values for which the equation has two distinct real roots.

Answer

We need to satisfy the condition $b^2 - 4ac > 0$ to have two distinct real roots.

The coefficients of the quadratic equation can be labelled $a = k, b = 4k \text{ and } c = 3k$, leading to

$$(4k)^2 - 4(k)(3k) > 0$$

$$16k^2 - 12k^2 > 0$$

$$4k^2 > 0$$

Thus we get the range of k as

$$k > 0$$

1.7. Simultaneous equations

A simultaneous equation is a **set of equations**, with **unknown variables**, that satisfy a condition.

These can be solved through the use of the methods of **elimination** and **substitution**.

Elimination

The method of elimination requires you to multiply one, or both, equation such that one of the unknown variables **shares** the **same coefficient** in both equations.

- This may require you to multiply the equations with an integer, or an unknown variable in more complicated problems.

Example

Solve the simultaneous equations

$$4a + 2b = 8$$

$$a + 3b = 7$$

Answer

- First we make an unknown variable have the same coefficient.
  - We can do this by multiplying the second equation by 4, resulting in $4a + 12b = 28$, making the unknown variable a share the same coefficient in both equations.

$$4a + 2b = 8$$

$$4a + 12b = 28$$

- Now we can subtract both equations to get rid of one unknown variable, resulting in $2b - 12b = 8 - 28$. This will help us isolate one variable and find its value.

$$-10b = -20$$

$$b = 2$$

- Substitute the value of b into any of the two equations. This is done to find the value of the other unknown variable.

$$4a + 2(2) = 8$$

$$a = 1$$

Substitution

The second method, substitution, makes one unknown variable the **subject** of the equation. This is then substituted into the second equation to find its value.

- Upon finding the value of one unknown variable, we can easily find the value of the other by substituting its value into any of the two equations.

Example

Solve the simultaneous equation

$$4a + 2b = 8$$

$$a + 3b = 7$$

Answer

- Using the second equation, we make a the subject of the formula.

$$a + 3b = 7 \implies a = 7 - 3b$$

- Substitute this expression into the first equation, $4a + 2b = 8$.

$$4(7 - 3b) + 2b = 8$$

$$2b - 12b = 8 - 28$$

$$b = 2$$

- Substitute this value of b into any of the two equations. This is done to find the value of the other unknown variable.

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$$a = 7 - 3(2)$$

$$a = 1$$

1.8. Simultaneous equations with quadratics

- A simultaneous equation can include a quadratic curve and a linear line.
  - Solving these can be thought of as finding the **points of intersection** of both equations.

Equating

One way of solving simultaneous equations is by equating both curves, or curve and line. This will result in a quadratic equation which can be solved to find the points of intersection.

Example

Find the coordinates of the intersections of the curve $y^2 = 4x^2 - 7$ and line $x + y = 5$ by equating both equations.

Answer

- Make y the subject of the formula in the line equation, and square both sides to get an equation in terms of y^2.

$$y = 5 - x \implies y^2 = (5 - x)^2$$

- Now we can equate both equations, as their points of intersections must have the same y coordinate.

$$(5 - x)^2 = 4x^2 - 7$$

$$x^2 - 10x + 25 = 4x^2 - 7$$

$$0 = 3x^2 + 10x - 32$$

We can solve this quadratic equation in terms of x, to get the x coordinates of intersection.

$$3x^2 + 10x - 32 \implies (3x + 16)(x - 2)$$

$$x = -\frac{16}{3} \text{ or } x = 2$$

- Substitute this value of x into the linear equation to find the y coordinates of intersection.

$$-\frac{16}{3} + y = 5 \implies y = \frac{31}{3}$$

$$2 + y = 5 \implies y = 3$$

- The solutions to this simultaneous equation, or points of intersection of both equations, is:

$$\left(-\frac{16}{3}, \frac{31}{3}\right) \text{ and } (2, 3)$$

Substitution

Another way of solving simultaneous equations is through the use of a substitution. This allows us to write an equation of one curve in terms of x or y. The new expression can then be substituted into the second equation to get the points of intersection.

Example

Find the coordinates of the intersections of the curve $y^2 = 4x^2 - 7$ and line $x + y = 5$ using a substitution.

Answer

- Make x the subject of the formula in the line equation to get $x = 5 - y$. Now we can substitute this expression as x in the quadratic equation.

$$y^2 = 4(5 - y)^2 - 7$$

$$y^2 = 100 - 40y + 4y^2 - 7$$

$$3y^2 - 40y + 93 = 0$$

We can solve this quadratic equation in terms of y, to get the y coordinates of intersection.

$$3y^2 - 40y + 93 \implies (3y - 31)(y - 3)$$

$$y = \frac{31}{3} \text{ or } y = 3$$

- Substitute this value of y into the linear equation to find the x coordinates of intersection.

$$x + \frac{31}{3} = 5 \implies x = -\frac{16}{3}$$

$$x + 3 = 5 \implies x = 2$$

- The solutions to this simultaneous equation, or points of intersection of both equations, is:

$$\left(-\frac{16}{3}, \frac{31}{3}\right) \text{ and } (2, 3)$$

1.9. Substitutions

Many questions may ask you to solve unique equations, which may not resemble a quadratic at first glance. These can be solved by making a clever substitution, by the use of a **variable**, and then factorising it to solve for the values of x.

The table below shows an **example** of possible substitutions that one can use

Equation	Substitution
$ax^4 + bx^2 + c$	$u = x^2$ and $u^2 = x^4$
$ax^6 + bx^3 + c$	$u = x^3$ and $u^2 = x^6$
$ax + b\sqrt{x} + c$	$u = \sqrt{x}$ and $u^2 = x$
$ax^{\frac{2}{3}} + bx^{\frac{1}{3}} + c$	$u = x^{\frac{1}{3}}$ and $u^2 = x^{\frac{2}{3}}$
$ax^{-2} + bx^{-1} + c$	$u = x^{-1}$ and $u^2 = x^{-2}$

Example 1

Solve the equation $x^4 - 5x^2 + 4 = 0$

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Answer

Let $u = x^2$, such that $u^2 = x^4$ and $x = \pm\sqrt{u}$

Using this substitution, we get $u^2 - 5u + 4 = 0$. This quadratic equation can be factorised easily.

$$(u-4)(u-1) = 0$$

$$u = 4 \implies x^2 = 4 \implies x = \pm 2$$

$$u = 1 \implies x^2 = 1 \implies x = \pm 1$$

Example 2

Solve the equation $6x + \sqrt{x} - 1 = 0$

Answer

Let $u = \sqrt{x}$, such that $u^2 = x$ and $x = u^2$

Using this substitution, we get $6u^2 + u - 1 = 0$. This quadratic can be factorised easily.

$$(3u-1)(2u+1) = 0$$

$$u = \frac{1}{3} \implies \sqrt{x} = \frac{1}{3} \implies x = \frac{1}{9}$$

$$u = \frac{-1}{2} \implies \sqrt{x} = \frac{-1}{2} \implies x = \frac{1}{4}$$

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